

Exhibit CC-143

DDR5 SDRAM RDIMM Addendum

MTC20F208XS1RC – 48GB 24Gb Die Revision B

Features

Information provided here is in addition to or supercedes information provided in the Micron DDR5 RDIMM Core data sheet.

- DDR5 functionality and operations supported as defined in the component data sheet
- Features and specifications defined in the Micron DDR5 RDIMM core data sheet
- 288-pin, DDR5 registered dual in-line memory module (DDR5 RDIMM)
- Fast data transfer rate: PC5-4800, PC5-5600, PC5-6400
- 48GB (6Gig x 80)
- Dual-rank
- 32 internal banks; 8 groups of 4 banks each

Figure 1: 288-Pin DDR5 RDIMM (R/C-E0)

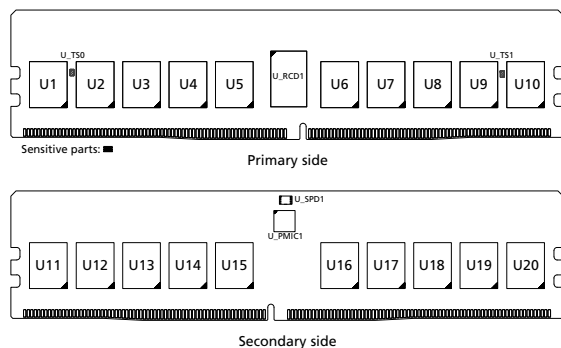
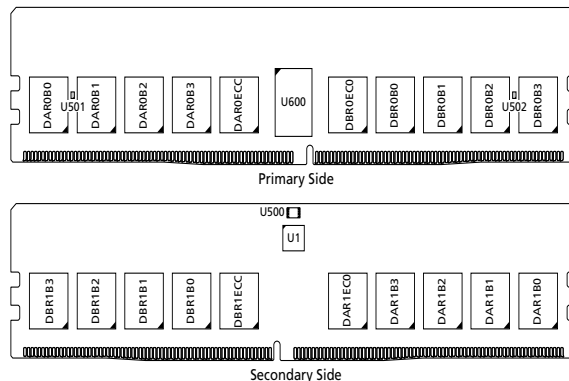


Figure 2: 288-Pin DDR5 RDIMM (R/C-E1)



Options

- Operating temperature
 - Commercial ($0^{\circ}\text{C} \leq T_{\text{OPER}} \leq 95^{\circ}\text{C}$)
- Frequency/CAS latency
 - 0.416ns @ CL = 40 (DDR5-4800)
 - 0.357ns @ CL = 46 (DDR5-5600)
 - 0.312ns @ CL = 52 (DDR5-6400)

Marking

C
48B
56B
64B



48GB (x80, ECC, DR) 288-Pin DDR5 RDIMM Features

Table 1: Addressing

Parameter	48GB
Row address ¹	96K (R0-R16) ²
Column address ¹	1K (C0-C9)
Device bank group address ¹	8 (BG0-BG2)
Device bank address per bank group ¹	4 (BA0-BA1)
Device configuration	24Gb (3Gb x 8), 32 banks
Module rank address	2 (CS0_n, CS1_n)

- Notes: 1. These parameters represent the logical address state of the CA bus for different commands. Refer to the command truth table in the component data sheet.
2. For non-binary densities, a quarter of the row address space is invalid. When the MSB address bit is HIGH, the MSB-1 address shall be LOW.

Table 2: Part Numbers and Timing Parameters – 48GB Modules

Base device: MT60B3G8,¹ 24Gb DDR5 SDRAM Die Revision B

Part Number	Module Density	Configuration	Module Bandwidth	Memory Clock/ Data Rate	Clock Cycles (CL-nRCD-nRP)	Designation ²
MTC20F208XS1RC48BB1	48GB	6Gb x 80	38.4 GB/s	0.416ns/4800 MT/s	40-39-39	Production
MTC20F208XS1RC56BB1	48GB	6Gb x 80	44.8 GB/s	0.357ns/5600 MT/s	46-45-45	Preliminary
MTC20F208XS1RC64BB1	48GB	6Gb x 80	51.2 GB/s	0.312ns/6400 MT/s	52-52-52	Advance

- Notes: 1. The data sheet for the base device can be found on [micron.com](https://www.micron.com).
2. **Production:** Although considered final, these specifications are subject to change as further product development and data characterization sometimes occur. **Preliminary:** For evaluation and reference purposes only and are subject to change by Micron without notice. Products are only warranted by Micron to meet Micron's production data sheet specifications. **Advance:** Contains initial descriptions of products still under development. For evaluation purposes only and are subject to change by Micron without notice. Products are only warranted by Micron to meet Micron's production data sheet specification.

Important Notes and Warnings

Micron Technology, Inc. ("Micron") reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions. This document supersedes and replaces all information supplied prior to the publication hereof. You may not rely on any information set forth in this document if you obtain the product described herein from any unauthorized distributor or other source not authorized by Micron.

Automotive Applications. Products are not designed or intended for use in automotive applications unless specifically designated by Micron as automotive-grade by their respective data sheets. Distributor and customer/distributor shall assume the sole risk and liability for and shall indemnify and hold Micron harmless against all claims, costs, damages, and expenses and reasonable attorneys' fees arising out of, directly or indirectly, any claim of product liability, personal injury, death, or property damage resulting directly or indirectly from any use of non-automotive-grade products in automotive applications. Customer/distributor shall ensure that the terms and conditions of sale between customer/distributor and any customer of distributor/customer (1) state that Micron products are not designed or intended for use in automotive applications unless specifically designated by Micron as automotive-grade by their respective data sheets and (2) require such customer of distributor/customer to indemnify and hold Micron harmless against all claims, costs, damages, and expenses and reasonable attorneys' fees arising out of, directly or indirectly, any claim of product liability, personal injury, death, or property damage resulting from any use of non-automotive-grade products in automotive applications.

Critical Applications. Products are not authorized for use in applications in which failure of the Micron component could result, directly or indirectly in death, personal injury, or severe property or environmental damage ("Critical Applications"). Customer must protect against death, personal injury, and severe property and environmental damage by incorporating safety design measures into customer's applications to ensure that failure of the Micron component will not result in such harms. Should customer or distributor purchase, use, or sell any Micron component for any critical application, customer and distributor shall indemnify and hold harmless Micron and its subsidiaries, subcontractors, and affiliates and the directors, officers, and employees of each against all claims, costs, damages, and expenses and reasonable attorneys' fees arising out of, directly or indirectly, any claim of product liability, personal injury, or death arising in any way out of such critical application, whether or not Micron or its subsidiaries, subcontractors, or affiliates were negligent in the design, manufacture, or warning of the Micron product.

Customer Responsibility. Customers are responsible for the design, manufacture, and operation of their systems, applications, and products using Micron products. ALL SEMICONDUCTOR PRODUCTS HAVE INHERENT FAILURE RATES AND LIMITED USEFUL LIVES. IT IS THE CUSTOMER'S SOLE RESPONSIBILITY TO DETERMINE WHETHER THE MICRON PRODUCT IS SUITABLE AND FIT FOR THE CUSTOMER'S SYSTEM, APPLICATION, OR PRODUCT. Customers must ensure that adequate design, manufacturing, and operating safeguards are included in customer's applications and products to eliminate the risk that personal injury, death, or severe property or environmental damages will result from failure of any semiconductor component.

Limited Warranty. In no event shall Micron be liable for any indirect, incidental, punitive, special or consequential damages (including without limitation lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort, warranty, breach of contract or other legal theory, unless explicitly stated in a written agreement executed by Micron's duly authorized representative.

DQ Map

Table 3: Component-to-Module DQ Map (R/C-E0)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
U1	0	3A	154	U2	0	11A	165
	1	0A	7		1	8A	18
	2	1A	9		2	9A	20
	3	2A	152		3	10A	163
	4	7A	161		4	15A	172
	5	4A	14		5	12A	25
	6	5A	16		6	13A	27
	7	6A	159		7	14A	170
U3	0	19A	176	U4	0	27A	187
	1	16A	29		1	24A	40
	2	17A	31		2	25A	42
	3	18A	174		3	26A	185
	4	23A	183		4	31A	194
	5	20A	36		5	28A	47
	6	21A	38		6	29A	49
	7	22A	181		7	30A	192
U5	0	CB3A	198	U6	0	CB3B	243
	1	CB0A	51		1	CB0B	96
	2	CB1A	53		2	CB1B	98
	3	CB2A	196		3	CB2B	241
	4	CB7A	205		4	CB7B	236
	5	CB4A	58		5	CB4B	89
	6	CB5A	60		6	CB5B	91
	7	CB6A	203		7	CB6B	234
U7	0	3B	247	U8	0	11B	258
	1	0B	100		1	8B	111
	2	1B	102		2	9B	113
	3	2B	245		3	10B	256
	4	7B	254		4	15B	265
	5	4B	107		5	12B	118
	6	5B	109		6	13B	120
	7	6B	252		7	14B	263

Table 3: Component-to-Module DQ Map (R/C-E0) (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
U9	0	19B	269	U10	0	27B	280
	1	16B	122		1	24B	133
	2	17B	124		2	25B	135
	3	18B	267		3	26B	278
	4	23B	276		4	31B	287
	5	20B	129		5	28B	140
	6	21B	131		6	29B	142
	7	22B	274		7	30B	285
U11	0	24B	133	U12	0	16B	122
	1	27B	280		1	19B	269
	2	26B	278		2	18B	267
	3	25B	135		3	17B	124
	4	28B	140		4	20B	129
	5	31B	287		5	23B	276
	6	30B	285		6	22B	274
	7	29B	142		7	21B	131
U13	0	8B	111	U14	0	0B	100
	1	11B	258		1	3B	247
	2	10B	256		2	2B	245
	3	9B	113		3	1B	102
	4	12B	118		4	4B	107
	5	15B	265		5	7B	254
	6	14B	263		6	6B	252
	7	13B	120		7	5B	109
U15	0	CB0B	96	U16	0	CB0A	51
	1	CB3B	243		1	CB3A	198
	2	CB2B	241		2	CB2A	196
	3	CB1B	98		3	CB1A	53
	4	CB4B	89		4	CB4A	58
	5	CB7B	236		5	CB7A	205
	6	CB6B	234		6	CB6A	203
	7	CB5B	91		7	CB5A	60



48GB (x80, ECC, DR) 288-Pin DDR5 RDIMM DQ Map

Table 3: Component-to-Module DQ Map (R/C-E0) (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
U17	0	24A	40	U18	0	16A	29
	1	27A	187		1	19A	176
	2	26A	185		2	18A	174
	3	25A	42		3	17A	31
	4	28A	47		4	20A	36
	5	31A	194		5	23A	183
	6	30A	192		6	22A	181
	7	29A	49		7	21A	38
U19	0	8A	18	U20	0	0A	7
	1	11A	165		1	3A	154
	2	10A	163		2	2A	152
	3	9A	20		3	1A	9
	4	12A	25		4	4A	14
	5	15A	172		5	7A	161
	6	14A	170		6	6A	159
	7	13A	27		7	5A	16

Table 4: Component-to-Module DQ Map (R/C-E1)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
DAR0B0	0	7A	161	DAR0B1	0	15A	172
	1	4A	14		1	12A	25
	2	5A	16		2	13A	27
	3	6A	159		3	14A	170
	4	3A	154		4	11A	165
	5	0A	7		5	8A	18
	6	1A	9		6	9A	20
	7	2A	152		7	10A	163
DAR0B2	0	19A	176	DAR0B3	0	27A	187
	1	16A	29		1	24A	40
	2	17A	31		2	25A	42
	3	18A	174		3	26A	185
	4	23A	183		4	31A	194
	5	20A	36		5	28A	47
	6	21A	38		6	29A	49
	7	22A	181		7	30A	192
DAR0ECC	0	CB3A	198	DAR1B0	0	4A	14
	1	CB0A	51		1	7A	161
	2	CB1A	53		2	6A	159
	3	CB2A	196		3	5A	16
	4	CB7A	205		4	0A	7
	5	CB4A	58		5	3A	154
	6	CB5A	60		6	2A	152
	7	CB6A	203		7	1A	9
DAR1B1	0	12A	25	DAR1B2	0	16A	29
	1	15A	172		1	19A	176
	2	14A	170		2	18A	174
	3	13A	27		3	17A	31
	4	8A	18		4	20A	36
	5	11A	165		5	23A	183
	6	10A	163		6	22A	181
	7	9A	20		7	21A	38

Table 4: Component-to-Module DQ Map (R/C-E1) (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
DAR1B3	0	24A	40	DAR1ECC	0	CB0A	51
	1	27A	187		1	CB3A	198
	2	26A	185		2	CB2A	196
	3	25A	42		3	CB1A	53
	4	28A	47		4	CB4A	58
	5	31A	194		5	CB7A	205
	6	30A	192		6	CB6A	203
	7	29A	49		7	CB5A	60
DBR0B0	0	7B	254	DBR0B1	0	15B	265
	1	4B	107		1	12B	118
	2	5B	109		2	13B	120
	3	6B	252		3	14B	263
	4	3B	247		4	11B	258
	5	0B	100		5	8B	111
	6	1B	102		6	9B	113
	7	2B	245		7	10B	256
DBR0B2	0	23B	276	DBR0B3	0	27B	280
	1	20B	129		1	24B	133
	2	21B	131		2	25B	135
	3	22B	274		3	26B	278
	4	19B	269		4	31B	287
	5	16B	122		5	28B	140
	6	17B	124		6	29B	142
	7	18B	267		7	30B	285
DBR0ECC	0	CB3B	243	DBR1B0	0	4B	107
	1	CB0B	96		1	7B	254
	2	CB1B	98		2	6B	252
	3	CB2B	241		3	5B	109
	4	CB7B	236		4	0B	100
	5	CB4B	89		5	3B	247
	6	CB5B	91		6	2B	245
	7	CB6B	234		7	1B	102

Table 4: Component-to-Module DQ Map (R/C-E1) (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
DBR1B1	0	12B	118	DBR1B2	0	20B	129
	1	15B	265		1	23B	276
	2	14B	263		2	22B	274
	3	13B	120		3	21B	131
	4	8B	111		4	16B	122
	5	11B	258		5	19B	269
	6	10B	256		6	18B	267
	7	9B	113		7	17B	124
DBR1B3	0	24B	133	DBR1ECC	0	CB0B	96
	1	27B	280		1	CB3B	243
	2	26B	278		2	CB2B	241
	3	25B	135		3	CB1B	98
	4	28B	140		4	CB4B	89
	5	31B	287		5	CB7B	236
	6	30B	285		6	CB6B	234
	7	29B	142		7	CB5B	91

I_{DD} Specifications

Table 5: DDR5 I_{DD} Specifications and Conditions – 48GB (Die Revision B)

Module I_{DD} is based on PMIC VIN_BULK 12V input current and typical operating range of temperature. Each I_{DD} parameter includes PMIC efficiency, RCD current and all DRAM current on all supplies (V_{DD}, V_{DDQ}, and V_{PP}).

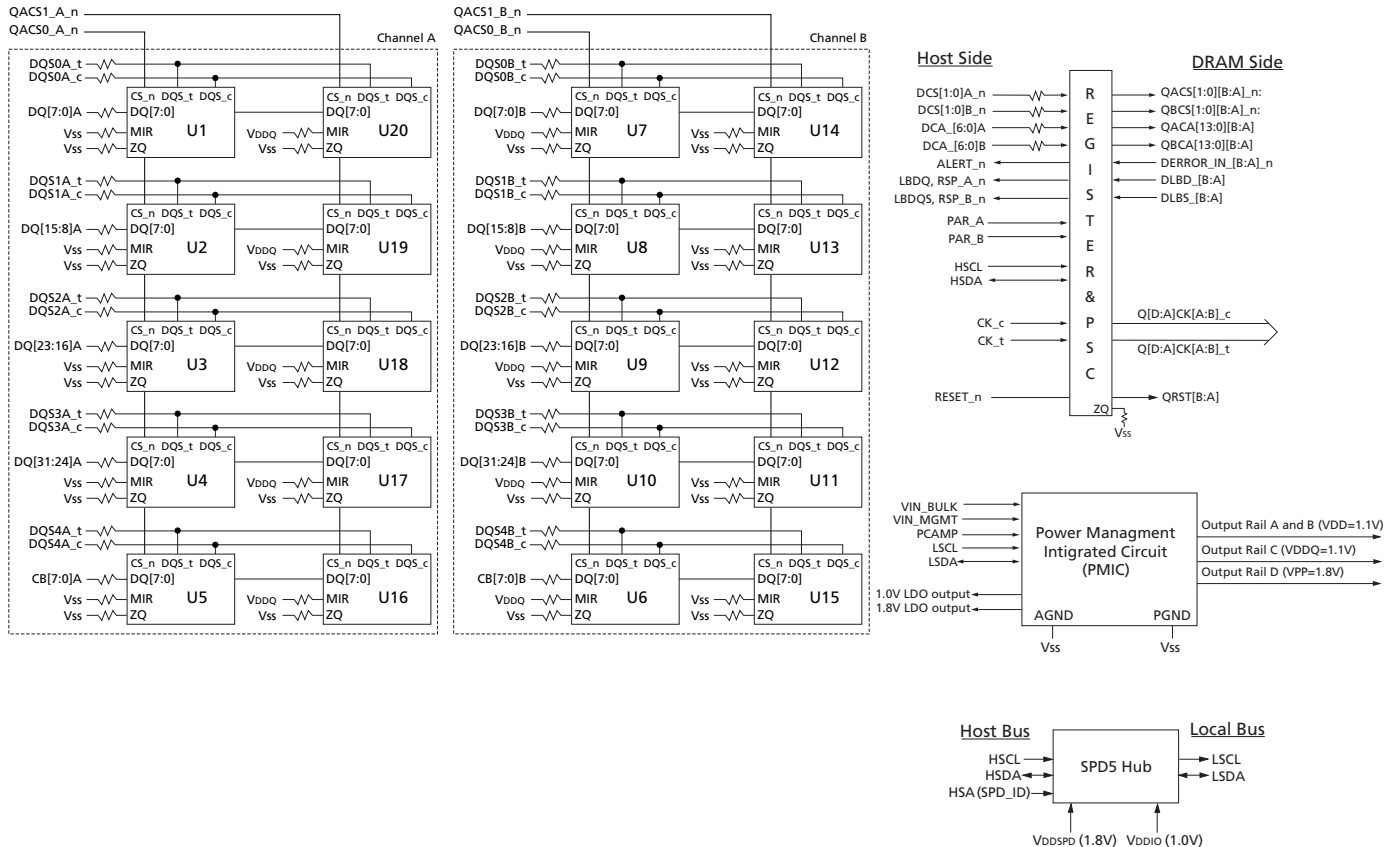
Parameter	Symbol	4800	5600	6400	Units
Operating one bank ACTIVATE-PRECHARGE current	I _{DD0} ¹	255	TBD	TBD	mA
Operating four bank ACTIVATE-PRECHARGE current	I _{DD0F} ¹	297	TBD	TBD	mA
Precharge standby current	I _{DD2N} ²	253	TBD	TBD	mA
Precharge standby non-target command	I _{DD2NT} ¹	286	TBD	TBD	mA
Precharge power-down current	I _{DD2P} ²	245	TBD	TBD	mA
Active standby current	I _{DD3N} ²	262	TBD	TBD	mA
Active power-down current	I _{DD3P} ²	267	TBD	TBD	mA
Operating burst read current	I _{DD4R} ¹	565	TBD	TBD	mA
Operating burst write current	I _{DD4W} ¹	665	TBD	TBD	mA
Operating burst write with write CRC current	I _{DD4WC} ¹	645	TBD	TBD	mA
Burst refresh (normal refresh mode) current	I _{DD5B} ¹	471	TBD	TBD	mA
Burst refresh (fine granularity refresh mode) current	I _{DD5F} ¹	461	TBD	TBD	mA
Burst refresh (same bank refresh mode) current	I _{DD5C} ¹	306	TBD	TBD	mA
Self refresh current	I _{DD6N} ²	159	TBD	TBD	mA
Operating bank interleave read current	I _{DD7} ¹	696	TBD	TBD	mA
Maximum power saving deep power down mode current	I _{DD8} ²	213	TBD	TBD	mA

Notes: 1. One module rank in this I_{DD}/I_{DDQ}/I_{PP} condition, the other rank in I_{DD2N}/I_{DDQ2N}/I_{PP2N}.

2. Both ranks in this I_{DD}/I_{DDQ}/I_{PP} condition.

Functional Block Diagram

Figure 3: Functional Block Diagram



- Notes: 1. The ZQ ball on each DDR5 component is connected to an external $240\Omega \pm 1\%$ resistor that is tied to ground. It is used for the calibration of the component's ODT and output driver.
2. Functional block diagram is for reference only.



Revision History

Rev. C – 01/2023

- Added IDD limits for 48B, and changed part status for 48B to "Production"
- Added R/C-E1 drawing to Features and R/C-E1 table to DQ Map
- Removed Micron Confidential marking
- Changed data sheet status to Production

Rev. B – 11/2022

- Added MIR state to Functional Block Diagram

Rev. A – 04/2022

- Preliminary Release

8000 S. Federal Way, P.O. Box 6, Boise, ID 83707-0006
208-368-4000, micron.com/support

Micron and the Micron logo are trademarks of Micron Technology, Inc.
All other trademarks are the property of their respective owners.

This data sheet contains minimum and maximum limits specified over the power supply and temperature range set forth herein. Although considered final, these specifications are subject to change, as further product development and data characterization sometimes occur.